

Instructions: -

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **NINE** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only):

[2 x 7 = 14]

- (a) The unit responsible for fetching, decoding, and executing instructions is
 - (i) ALU
 - (ii) Control Unit
 - (iii) Memory Unit
 - (iv) I/O Unit
- (b) Which addressing mode uses the contents of a register to determine the effective address?
 - (i) Signed integers
 - (ii) Indirect addressing
 - (iii) Direct addressing
 - (iv) Register addressing
- (c) Two's complement representation is primarily used for
 - (i) Signed integers
 - (ii) ASCII characters
 - (iii) Instruction opcodes
 - (iv) Floating point numbers
- (d) Booth's algorithm is used for
 - (i) Floating point operations
 - (ii) Division
 - (iii) Character coding
 - (iv) Multiplication
- (e) In floating point representation, the part representing significant digits is called
 - (i) Mantissa
 - (ii) Exponent
 - (iii) Sign bit
 - (iv) Bias
- (f) x86 architecture belongs to which processor family?
 - (i) RISC
 - (ii) EPIC
 - (iii) CISC
 - (iv) ARM
- (g) Memory chips used in main memory are typically
 - (i) Optical memory
 - (ii) Magnetic tapes
 - (iii) Flash
 - (iv) DRAM
- (h) USB is an example of
 - (i) Networking device
 - (ii) Standard I/O interface
 - (iii) Storage device only
 - (iv) Wireless protocol
- (i) Pipeline speedup ideally increases with
 - (i) Number of control units
 - (ii) Number of pipeline stages
 - (iii) Size of RAM
 - (iv) Size of cache
- (j) In cache memory, FIFO and LRU are examples of
 - (i) Mapping techniques
 - (ii) Replacement algorithms
 - (iii) Addressing techniques
 - (iv) Write policies

- Q.2** (a) Explain the functional blocks of a computer with neat diagrams: CPU, Memory, and I/O subsystems. **[7]**
- (b) Explain addressing modes with examples. Compare at least five different addressing techniques. **[7]**

Bihar Engineering University, Patna
B.Tech 4th Semester Examination, 2024



Course : B.Tech
Code : 105401

Subject: Computer Organization & Architecture

Time: 03 Hours
Full Marks: 70

Instructions:-

- (i) The marks are indicated in the right-hand margin
- (ii) There are **NINE** questions in this paper
- (iii) Attempt **FIVE** questions in all
- (iv) Question No. 1 is compulsory

Q.1 Choose the correct answer of the following (any seven question only):

[3 x 7 = 14]

- (a) What is the first step in the instruction execution cycle?
 - (i) Decode
 - (ii) Execute
 - ☒ (iii) Fetch
 - (iv) Store
- (b) What is the advantage of carry look-ahead adders over ripple carry adders?
 - (i) Simplicity in design
 - (ii) Increased latency
 - (iii) Reduced power consumption
 - ☒ (iv) Faster addition by reducing carry propagation delay
- (c) Which type of control unit is faster but more difficult to modify?
 - ☒ (i) Hardwired Control Unit
 - (ii) Microprogrammed Control Unit
 - (iii) Cache Control Unit
 - (iv) Memory Management Unit
- (d) Which of the following I/O transfer methods involves the CPU polling the device until data is ready?
 - (i) DMA
 - (ii) Interrupt-driven I/O
 - (iii) Program-controlled I/O
 - (iv) Asynchronous I/O
- (e) What does a branch prediction mechanism in a CPU help reduce?
 - (i) Data hazards
 - (ii) Structural hazards
 - ☒ (iii) Control hazards
 - (iv) Clock cycle delays
- (f) Which of the following is NOT a type of parallel processor architecture?
 - (i) SIMD
 - ☒ (ii) SISO
 - (iii) MIMD
 - (iv) Multi-core
- (g) Which cache replacement algorithm removes the least recently used block?
 - (i) FIFO
 - (ii) LFL
 - ☒ (iii) LRL
 - (iv) Random Replacement
- (h) Which technique is used to handle data hazards in pipelining?
 - (i) Increasing memory size
 - ☒ (ii) Instruction scheduling
 - (iii) Parallel processing
 - (iv) Increasing clock speed
- (i) _____ addressing mode directly specifies the operand within the instruction?
 - (i) Direct Addressing
 - ☒ (ii) Immediate Addressing
 - (iii) Indirect Addressing
 - (iv) Indexed Addressing



- (j) A computer with 32 bit wide data bus uses 1K X 8 static RAM memory chips. What is smallest memory that this computer have?
(i) 32K (ii) 16K
(iii) 8K (iv) 24K

Q.2 (a) Describe the Instruction execution cycle in a CPU. Explain some key features of the x86 architecture [7]

(b) Explain different addressing modes in a CPU with an examples. [7]

Q.3 (a) Explain the Addition and Subtraction Algorithm with the help of flowchart. [7]
(b) What is a Carry-look ahead fast adder? Draw the circuit of an 8-bit carry-look ahead adder using 4-bit carry look ahead adder. [7]

Q.4 (a) Explain different cache replacement algorithms used in memory management. [7]

(b) What are the different types of Registers? With the diagram show how the basic computer register connected to a common Bus. [7]

Q.5 (a) Illustrate what are the challenges associated with concurrent access to memory in parallel processing? [7]

(b) Describe different cache coherence protocols used in parallel processors [7]

Q.6 (a) A non-pipeline system takes 50ns to process a task. The same task can be processed in a six-segment pipeline with a clock cycle of 10ns. Determine the speedup ratio of the pipeline for 100 task. [7]

(b) What are pipeline hazards? Explain different types of hazards in pipelining [7]

Q.7 Explain Booth's algorithm for multiplication with the help of flowchart. Provide an example for its implementation. [14]

Q.8 Describe different types of Input/Output transfers. Explain how data transfer take place through DMA. [14]

Q.9 Write short notes on any two of the following:- [2 x 2 = 4]

- (a) RISC and CISC processor
- (b) fixed and floating point representations
- (c) I/O device interfaces
- (d) Instruction Set Architecture





Bihar Engineering University, Patna
End Semester Examination – 2023

Course: B.Tech.

Code: 105401

Semester-IV

Subject: Computer Organization and Architecture

Time: 03 Hours

Full Marks: 70

Instructions:-

- (i) The marks are indicated in the right-hand margin.
- (ii) There are NINE questions in this paper.
- (iii) Attempt FIVE questions in all.
- (iv) Question No. 1 is compulsory.

Q.1 Choose the correct option of the following (any seven only):

[2 x 7 = 14]

- (a) A pipeline stage
 - (i) Is sequential circuit
 - (ii) Is combination circuit
 - (iii) Consists of both sequential and combinational circuit
 - (iv) None of these
- (b) A direct mapped cache memory with n blocks is nothing but which of the following set associative cache memory originations

(i) 0-way set associative	(ii) 1-way set associative
(iii) 2-way set associative	(iv) n-way set associative
- (c) The performance of a pipelined processor suffers if
 - (i) The pipeline stages have different delays
 - (ii) Consecutive instruction are dependent on each other
 - (iii) The pipeline stages share hardware resources
 - (iv) All of these
- (d) A computer with cache access time of 100 ns, a main memory access time of 1000 ns, and a hit ratio of 0.9 produces an average access time of

(i) 250 ns	(ii) 200 ns
(iii) 190 ns	(iv) None of these
- (e) Which of the following has no practical usage?

(i) SISD	(ii) SIMD
(iii) MISD	(iv) MIMD
- (f) A micro programmed control unit
 - (i) Is faster than a hardwired control unit
 - (ii) Facilitates easy implementation of new instructions
 - (iii) Is useful when every small program is to be run
 - (iv) Usually refers to the control unit of the microprocessor
- (g) In memory-mapped I/O.....
 - (i) The I/O devices and the memory share the same address space.
 - (ii) The I/O device have a separate address space
 - (iii) The memory and I/O device have an associated address space
 - (iv) A part of the memory is specifically set aside for the I/O operation
- (h) How many 128x8 bit RAMs are required to design 32 k x 32 bit RAM?

(i) 512	(ii) 128	(iii) 1024	(iv) 32
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- (i) The stalling of the processor due to the unavailability of the instruction is called as

(i) Control hazard	(ii) Structural hazard
(iii) Input hazard	(iv) None of the above
- (j) The addressing mode, where you directly specify the operand value is

(i) Immediate	(ii) Direct
(iii) Definite	(iv) Relative

Q.2

- (a) What are the hazards in pipeline architecture? Explain its types with suitable example. [7]
- (b) What is addressing mode? Why do computers use addressing mode techniques? Explain two modes with example, which do not use address fields. [7]



- ~~Q.4~~ (a) A 4-way set associative cache memory unit with a capacity of 16 KB is built using a block size of 8 words. The word length is 32 bits. The size of the physical address space is 4GB. Find the number of bits for TAG, SET, and WORD fields in the address generated by CPU. [7]
- (b) How is the virtual address mapped into physical address? What are the different methods of writing into cache? [7]
- ~~Q.5~~ (a) What are the different types of instruction formats? [7]
- (b) Discuss the different mapping techniques used in cache memories and their relative merits and demerits. [7]
- ~~Q.6~~ (a) Design a 4-bit carry-look ahead adder and explain its operation with an example. [7]
- (b) Consider a direct mapped cache with 8 cache blocks (numbered 0-7). If the memory block requests are in the following order 3, 5, 2, 8, 0, 63, 9, 16, 20, 17, 25, 18, 30, 24, 2, 63, 5, 82, 17, 24. What would be the status of cache blocks (block numbers residing in cache) at the end of the sequence. [7]
- ~~Q.7~~ (a) What is DMA? Describe how DMA is used to transfer data from peripherals. [7]
- (b) Differentiate between hardwired and micro programmed control unit. Explain each component of hardwired control unit organization. [7]
- ~~Q.8~~ (a) What do you mean by asynchronous data transfer? Explain strobe control and hand shaking mechanism. [7]
- (b) Show the systematic multiplication process of $(20) \times (-19)$ using Booth's algorithm [7]
- ~~Q.9~~ (a) The stage delays in a four stages pipeline are 800, 500, 400 and 300 picoseconds. The first stage (with delay 800 picoseconds) is replaced with a functionally equivalent design involving two stages with respective delays 600 and 350 picoseconds. What would be the throughput increases (in percentage) of the pipeline? [7]
- (b) Explain IEEE standard for floating point representation with example. [7]
- ~~Q.9~~ Write short notes on *any two* of the following : [7x2=14]
- Paging
 - Memory interleaving
 - Privileged and non-privileged instructions
 - Locality of reference
-



Code : 105401/106401

(2)

B.Tech 4th Semester Exam., 2022

(New Course)

**COMPUTER ORGANIZATION AND
ARCHITECTURE**

Time : 3 hours

Full Marks : 70

Instructions :

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **NINE** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question No. 1 is compulsory.

**1. Choose the correct answer of the following
(any seven) : $2 \times 7 = 14$**

- (a) The 8-bit encoding format used to store data in a computer is
 - (i) ASCII
 - (ii) EBCDIC
 - (iii) ANCI
 - (iv) USCII

(b) The bus used to connect the monitor to the CPU is

- (i) PCI bus
- (ii) SCSI bus
- (iii) memory bus
- (iv) ram bus

(c) The alternate way of writing the instruction, ADD #5, R1 is

- (i) ADD [5],[R1];
- (ii) ADDI 5,R1;
- (iii) ADDIME 5,[R1];
- (iv) There is no other way

(d) The instruction fetch phase ends with

- (i) placing the data from the address in MAR into MDR
- (ii) placing the address of the data into MAR
- (iii) completing the execution of the data and placing its storage address into MAR
- (iv) decoding the data in MDR and placing it in IR

AK23/320

(Turn Over)

AK23/320

(Continued)



(3)

(4)

(e) For converting a virtual address into the physical address, the programs are divided into

- (i) pages
- (ii) frames
- (iii) segments
- (iv) blocks

(f) The transfer of large chunks of data with the involvement of the processor is done by

- (i) DMA controller
- (ii) arbitrator
- (iii) user system programs
- (iv) None of the above

(g) The computer architecture aimed at reducing the time of execution of instructions is

- (i) CISC
- (ii) RISC
- (iii) ISA
- (iv) ANNA

(h) Which table handle stores the addresses of the interrupt handling sub-routines?

- (i) Interrupt-vector table
- (ii) Vector table
- (iii) Symbol link table
- (iv) None of the above

(i) The situation wherein the data of operands are not available is called

- (i) data hazard
- (ii) stock
- (iii) deadlock
- (iv) structural hazard

(j) The DMA controller has ____ registers.

- (i) 4
- (ii) 2
- (iii) 3
- (iv) 1

2. (a) List and briefly define the main structural components of a computer. 7

(b) Discuss the design and logic of a microprogram sequence. 7

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(Turn Over)

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(Continued)



(5)

3. Consider a hypothetical 32-bit microprocessor having 32-bit instructions composed of two fields—the first byte contains the opcode and the remainder the immediate operand or an operand address.

(a) What is the maximum directly addressable memory capacity (in bytes)?

(b) Discuss the impact on the system speed if the microprocessor bus has—

(i) a 32-bit local address bus and a 16-bit local data bus, or

(ii) a 16-bit local address bus and a 16-bit local data bus.

(c) How many bits are needed for the program counter and the instruction register?

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4. (a) A set-associative cache has a block size of four 16-bit words and a set size of 2. The cache can accommodate a total of 4096 words. The main memory size that is cacheable is 64K 32 bits. Design the cache structure and show how the processor's addresses are interpreted.

7

(b) Explain two techniques for enhancing the performance of computers with multiple execution pipelines.

7

(6)

5. (a) Calculate $(72530 - 13250)$ using ten's complement arithmetic. Assume rules similar to those for two's complement arithmetic.

7

(b) List and briefly explain five important instruction set design issues.

7

6. The x86 architecture includes an instruction called decimal adjust after addition (DAA). DAA performs the following sequence of instructions :

if $((AL \text{ AND } 0FH) > 9)$ OR $(AF = 1)$ then

AL $\leftarrow$ AL + 6;

AF $\leftarrow$ 1;

else

AF $\leftarrow$ 0;

endif;

if $(AL > 9FH)$ OR $(CF = 1)$ then

AL $\leftarrow$ AL + 60H;

CF $\leftarrow$ 1;

else

CF $\leftarrow$ 0;

endif.

"H" indicates hexadecimal. AL is an 8-bit register that holds the result of addition of two unsigned 8-bit integers. AF is a flag set if there is a carry from bit 3 to bit 4 in the result of an addition. CF is a flag set if there is a carry from bit 7 to bit 8. Explain the function performed by the DAA instruction.

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(Turn Over)

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(Continued)



(7)

7. A non-pipelined processor has a clock rate of 2.5 GHz and an average CPI (cycles per instruction) of 4. An upgrade to the processor introduces a five-stage pipeline. However, due to internal pipeline delays, such as latch delay, the clock rate of the new processor has to be reduced to 2 GHz.

(a) What is the speedup achieved for a typical program?

(b) What is the MIPS rate for each processor?

14

8. (a) Briefly explain the two basic approaches used to minimize register-memory operations on RISC machines. 7

(b) A computer has 16 registers, an ALU with 32 operations, and a shifter with 8 operations, all connected to a common bus system.

(i) Formulate a control word for micro-operation.

(ii) Show the bits of the control word that specify the micro-operation $R4 \leftarrow R5 + R6$.

7

(8)

9. Let α be the percentage of a program code that can be executed simultaneously by n processors in a computer system. Assume that the remaining code must be executed sequentially by a single processor. Each processor has an execution rate of x MIPS.

(a) Derive an expression for the effective MIPS rate when using the system for exclusive execution of this program, in terms of n , α and x .

(b) If $n = 16$ and $x = 4$ MIPS, determine the value of α that will yield a system performance of 40 MIPS.

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(Turn Over)

AK23—4600/320



Code : 051402

(2)

B.Tech 4th Semester Exam., 2019

COMPUTER ARCHITECTURE

Time : 3 hours

Full Marks : 70

Instructions :

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **NINE** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question No. 1 is compulsory.

1. Choose the correct answer for any seven of the following : 2×7=14

- (a) How many 128×8 bit RAMs are required to design $32 \text{ k} \times 32$ bit RAM?
 - (i) 512
 - (ii) 128
 - (iii) 1024
 - (iv) 32
- (b) The intradata transfer techniques are implemented using
 - (i) serial I/O
 - (ii) parallel I/O
 - (iii) Both (i) and (ii)
 - (iv) Neither (i) nor (ii)

(c) The sequence of events that happen during a typical fetch operation is

- (i) PC→MAR→Memory→MDR→IR
- (ii) PC→Memory→MDR→IR
- (iii) PC→Memory→IR
- (iv) PC→MAR→Memory→IR

(d) In case of pipelining processor, loop buffer is

- (i) very high speed memory maintained by the instruction fetch stage
- (ii) very high speed memory maintained by the instruction decode stage
- (iii) very high speed memory maintained by the instruction execute stage
- (iv) None of the above

(e) The average memory access time for a machine with a cache hit rate of 90% where the cache access time is 10 ns and the memory access time is 100 ns is

- (i) 55 ns
- (ii) 45 ns
- (iii) 90 ns
- (iv) 19 ns

AK9/711

(Turn Over)

AK9/711

(Continued)



(3)

(f) The minimum time delay between the initiations of two independent memory operations is called

- (i) access time
- ☒ (ii) cycle time
- (iii) transfer rate
- (iv) latency time

(g) In case of vectored interrupt, interrupt vector means

- (i) the branch information from the source which interrupts the system
- (ii) an address that points to a location in memory where the beginning address of the I/O service routine is stored

- ☒ (iii) Both (i) and (ii)
- (iv) None of the above

(h) A microprogrammed control unit

- (i) is faster than a hardwired control unit
- (ii) facilitates easy implementation of new instructions
- ☒ (iii) is useful when every small program is to be run
- (iv) usually refers to the control unit of the microprocessor

(i) Relative addressing mode is used to write position independent code because

- ☒ (i) the code in this mode is easy to atomize
- (ii) the code in this mode is easy to relocate in the memory
- (iii) the code in this mode is easy to make resident
- (iv) the code execution faster in this mode <http://www.akubihar.com>

(j) Which of the following holds data and processing instructions temporarily until the CPU needs it?

- (i) ROM
- (ii) Control unit
- ☒ (iii) Main memory
- (iv) Coprocessor chip

2. (a) How do instruction set, compiler technology, CPU implementation and control, and cache and memory hierarchy affect the CPU performance? Justify the effects in terms of program length, clock rate and effective CPI.

(b) How is virtual memory managed using paging and TLB? Explain with suitable example.

7+7=14

AK9/711

(Turn Over)

AK9/711

(Continued)



(5)

3. (a) Explain register reference and memory reference instructions in detail with suitable examples.
- (b) Draw the block diagram of control unit of basic computer. Explain in detail with control timing diagrams. $7+7=14$
4. (a) Explain one-, two- and three-address instruction with suitable examples.
- (b) Give an integrated diagram, showing the TLB and cache operations for a logical/virtual address generated by a processor. $7+7=14$
5. (a) Explain the daisy chaining mechanism for bus arbitration. Analyze the three bus arbitration methods—daisy chaining, polling and independent requesting with respect to communication reliability in the even of hardware failures.
- (b) Give the block diagram of microprogram sequencer for a control memory and explain it properly. $7+7=14$
6. (a) What do you understand by hardwired control? Give various methods to design hardwired control unit. Describe any one with suitable example.

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(Turn Over)

(6)

- (b) Describe autoincrement and auto-decrement addressing modes with proper examples. $7+7=14$
7. (a) What is direct memory access? Explain. Give block diagram of circuitry required for direct memory access.
- (b) A digital computer has a common bus system of 16 registers of 32 bits each. The bus is constructed with multiplexers.
- (i) How many selection inputs are there in each multiplexer?
- (ii) What size of multiplexers is needed? $7+7=14$
8. (a) When do you say the floating point number is normalized? Explain how floating point representation of number is done. Represent the number $(+46.25)$ as floating point binary number with 32 bits.
- (b) What are hazards in pipeline architecture? Explain its type with suitable example. $7+7=14$

AK9/711

(Continued)



(7)

9. (a) What is array processor? Explain SIMD array processor with suitable example.
- (b) A DMA controller transfers 16-bit words to memory using cycle stealing. The words are assembled from a device that transmits characters at the rate of 2400 characters per second. The CPU is fetching and executing instructions at an average rate of 1 million instructions per second. By how much will the CPU be slowed down because of DMA transfer? $7+7=14$

EDULOGY

Code : 051402



Code : 051402

B.Tech 4th Semester Examination, 2017

Computer Architecture

Time : 3 hours

Full Marks : 70

Instructions :

- (i) *There are Nine Questions in this Paper.*
 - (ii) *Attempt Five questions in all.*
 - (iii) **Question No. 1 is Compulsory.**
 - (iv) *The marks are indicated in the right-hand margin.*
- Each question carry in equal marks.*

1. Answer the following questions (any seven):

- (a) Write the range of decimal integer can be represented by n -bit 2's complement representation.
- (b) Justify the statement "Stack computer consists of an operation code only with no address field".
- (c) What do you mean by Arithmetic shift left operation?
- (d) What do you mean by locality of reference?
- (e) What are the properties of an ideal instruction set computer.
- (f) Define the term hardware polling.
- (g) What do you mean by data hazards in pipelining?

- (h) Explain indirect address mode, and how the effective address is calculated in this case.
- (i) Explain the use of subroutine with the help of suitable example.

- 2. Why is read and write control lines in a DMA controller bidirectional? Under what condition and for what purpose are they used as inputs?
- 3. Explain the concept of virtual memory with the help of diagram. Explain how virtual address is mapped to actual physical address.
- 4. What is meant by Addressing Mode? Explain at least five different Addressing Modes with an example.
- 5. What are the different conflicts that will arise in pipeline (elaborate)? How do you remove the conflicts?
- 6. Explain Von Neumann Architecture. What are its drawbacks?
- 7. What is a page fault? What does a page fault signify? Explain the different page replacement algorithms which determine the page to be removed in case of full memory.
- 8. Why does I/O interrupt make more efficient use of the CPU? Explain the sequence of operations that take place in an interrupt driven I/O transfer.

P.T.O.

Code : 051402

2



9. How many ROM chips are required to produce a memory capacity of 4096 bytes? How many address lines are required to access the 4096 bytes? How many of these addresses will be common to all these chips?

EDULOGY



Code : 051402

B.Tech 4th Semester Exam., 2016

COMPUTER ARCHITECTURE

Time : 3 hours

Full Marks : 70

Instructions :

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **NINE** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question No. 1 is compulsory.

1. Answer the following questions (any seven) :

2×7=14

- (a) Write the range of decimal integer can be represented by n -bit 1's complement representation.
- (b) What is zero address instruction format?
- (c) What do you mean by interrupt-initiated I/O?
- (d) What do you mean by cache coherence problem?

AK16/636

(Turn Over)



(2)

- (e) How many 128×8 RAM chips are needed to provide a memory capacity of 2048 bytes?
- (f) Briefly explain the conflicts in instruction pipeline.
- (g) Explain the use of subroutine with the help of suitable example.
- (h) What is a micro-operation? Explain the four categories of the most common micro-operations.
- (i) Define hit ratio and explain its significance.
2. With neat block diagram, explain how DMA controller is initialized for DMA data transfer. 14
3. Discuss Flynn's classification of computer. 14
4. What is associative memory? Explain with the help of a block diagram. Also mention the situation in which associative memory can be effectively utilized. 14
5. Discuss the different ways in which ROM can be programmed. 14

(3)

6. What is meant by pipelining? Why do we require instruction pipelining? Explain its working procedure. Discuss the pipeline performance measures. 14
7. A virtual memory system has an address space of 8 k words, memory space of 4 k words and page & block size of 1 k words. The following page reference changes occur during a given time interval :
4, 2, 0, 1, 2, 6, 1, 4, 0, 1, 0, 2, 3, 5, 7
Determine the four pages that are resident in main memory after each page reference change if the replacement algorithm used is (i) FIFO and (ii) LRU. 14
8. Explain the Strobe Control method of asynchronous data transfer. What are the disadvantages of this method? 14
9. Write short notes on the following : $7 \times 2 = 14$
(a) Shift instructions
(b) Daisy chaining priority

AK16/636

(Continued)

AK16-930/636

Code : 051402



Code : 051402

(2)

B.Tech. 4th Semester Exam., 2015

COMPUTER ARCHITECTURE

Time : 3 hours

Full Marks : 70

Instructions :

- (i) All questions carry equal marks.
- (ii) There are **NINE** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question No. 1 is compulsory.

1. Answer the following questions (any seven) :

- (a) What is the difference between Programmed I/O and Interrupt-driven I/O?
- (b) Explain different types of interrupts.
- (c) Define program counter.
- (d) Describe the phases in an instruction cycle.
- (e) What do you mean by direct memory access? akubihar.com
- (f) What is the difference between hardwired control and microprogrammed control?

(g) What is the importance of cache memory in a computer?

(h) Briefly explain subroutine call and return.

(i) What do you mean by array processors? How are they different from scalar processors?

- 2. Explain in detail the different addressing modes.
- 3. Explain in detail the hierarchical memory structure with a neat and labelled diagram.
- 4. What is swapping and how is it useful in memory management?
- 5. How does a DMA controller work? Explain with an example.
- 6. Define paging in memory management and write its advantages.
- 7. Explain in detail daisy-chain priority interrupt.
- 8. Explain in detail the three mapping procedures related to cache memory.
- 9. Explain in detail the handshake protocol of asynchronous data transfer.

AK15—940/564

(Turn Over)

AK15—940/564

Code : 051402



Code : 051402

(2)

B.Tech 4th Semester Exam., 2014

COMPUTER ARCHITECTURE

Time : 3 hours

Full Marks : 70

Instructions :

- (i) All questions carry equal marks.
- (ii) There are **NINE** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question No. 1 is compulsory.

1. Answer any seven of the following questions :

- (a) Registers R1 and R2 of a computer contain the decimal values 1200 and 4600. What is the effective address of the memory operand in each of the following instructions?
 - (i) Load 20(R1), R5
 - (ii) Add -(R2), R5
- (b) What is the use of condition code register?
- (c) What do you mean by end-around carry correction?
- (d) Discuss the role of Booth algorithm in the design of fast multipliers.

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- (e) Why is the wait-for-memory-function-completed step needed when reading from or writing to the main memory?

- (f) Write the sequence of control steps required for three-bus structure for the following instruction :

Add R4, R5, R6

- (g) Define locality of reference.
- (h) Give the features of a ROM cell.
- (i) What is the difference between a subroutine and an interrupt service routine?
- (j) Define bus arbitration.

- 2. (a) Explain the Flynn's classification of computer system architecture.

- (b) Explain MIPS and MFLOPS.

- 3. A program runs on a 10 GHz CPU with the instruction mix and corresponding clock cycle count 20 as given in the table :

Instruction type	Clock cycle count	Instruction count
Control transfer	5	5000
Data transfer	4	4000
Floating point	2	2000
Integer	1	1000

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(Continued)



(3)

Determine the following :

- (a) CPI
- (b) Execution time
- (c) MIPS rate for program

- 4. (a) Explain fetch decode execution cycle.
- (b) Differentiate between hardwired programmed control and microprogrammed control.
- 5. (a) Explain stack-based architecture of a CPU with the help of a diagram.
- (b) List registers of a non-pipelined CPU. Explain the purpose of each register.
- 6. (a) Differentiate between expansion and extension of memory.
- (b) Describe organization of a typical RAM chip. Differentiate between static RAM chip and dynamic RAM chip.
- 7. (a) Explain swapping. Why is it used in memory management?
- (b) Explain virtual memory. Explain the role of logical as well as physical address.

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- 8. Explain the use of DMA controllers in a computer system with a neat diagram.
- 9. Explain handshake protocol. Depict clearly how it controls data transfer during an input operation.

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